Arrays in Java

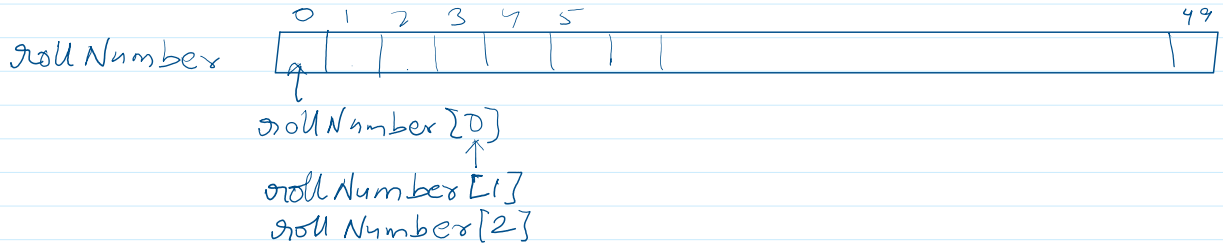
07 November 2024 19:57

WAP in Java to input roll number and marks of 50 students in a class and find the roll number of highest marks, find the roll number of lowest marks, find the average marks. And find all the roll numbers which are getting marks lower than average marks.

rollNumber ← 50

Marks ← 50

Marks1, Mark2, Mark3, Mark4 - - - - Mark50



In Java, Arrays are data structure that store multiple values of the same type in a single variable. They are useful for organizing collection of data allowing you to store a fixed size sequential collection of elements.

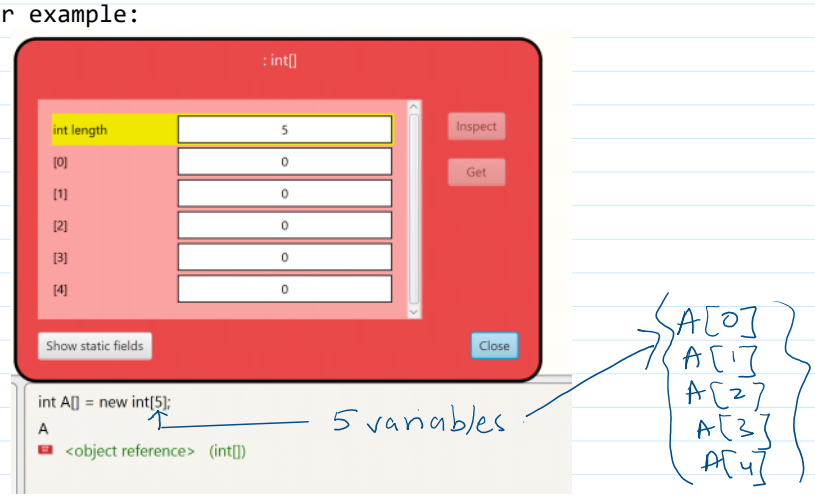
Arrays are collection of homogenous or similar data type elements. It occupies the memory in a sequence or in a contiguous form.

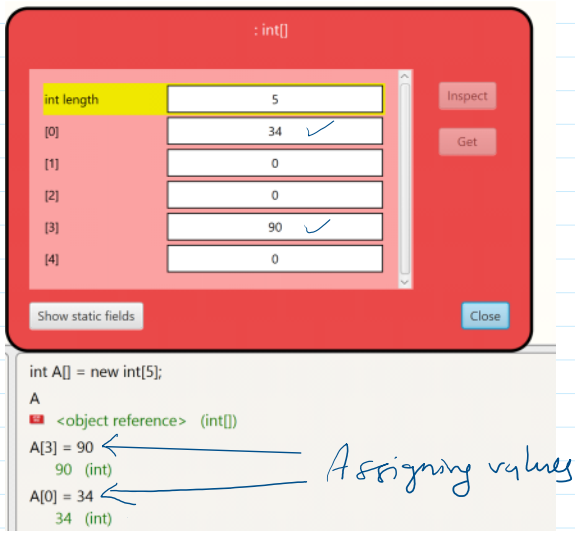
How to create array in Java?

Syntax:

```
data_type array_name[] = new data_type[number_of_elements];
```

For example:





Assigning values to array elements.

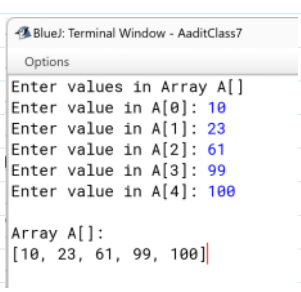
```
package ArraysInJava;
import java.util.*;


/**
 * Write a description of class One here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class One
{
    public static void main(String args[]){
        int A[] = new int[5];
        Scanner sc = new Scanner(System.in);
        int i;
        System.out.print("\fEnter values in Array A[] \n");
        for(i = 0; i < 5; i++){
            System.out.print("Enter value in A["+i+"]: ");
            A[i] = sc.nextInt();
        }
```

↑value of 'i' is changing, and the variable is changing

```
        System.out.print("\nArray A[]:\n[");
        for(i = 0; i < 4; i++){
            System.out.print(A[i]+", ");
        }
        System.out.print(A[i]+"]\n");
    }
}
```

Output:

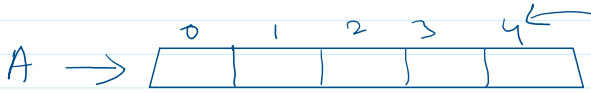


Array is a group of variable having same name and same data type. They reside on the memory in a sequence. The elements of an Array can be randomly accessed by using array-name along with subscript or index value.

A[i]

A[i]
↑ subscript or index value.

A subscript or index value of an array always starts from 0 (zero) and end at (N-1).
For example
int A[] = new int[5];

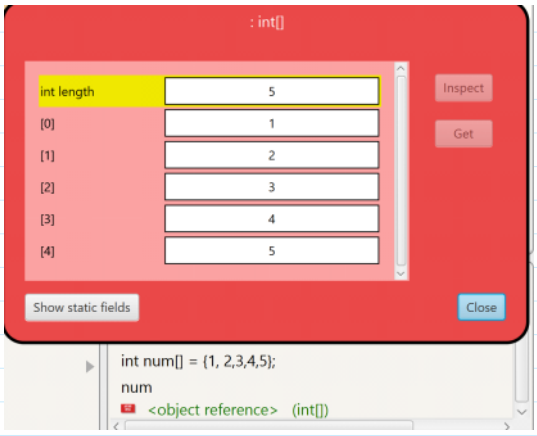


N-1
5-1

instantiation of an Array of 5 int's.

Initializing an Array:

int num[] = {1,2,3,4,5};




```
Accessing Array Elements:
      Array_name[valid_index_value];
   For example:
      int num[] = {1, 2,3,4,5};
      num
      <object reference>   (int[])
      num[0]
      1   (int)
      num[3]
      4   (int)
```

Array Length

- **The *length* property returns the size of the array(number of elements or variables in the array.**

  num.length
  5 (int) ← variable which stores number of elements in the array. And, it is very much useful in programs
-

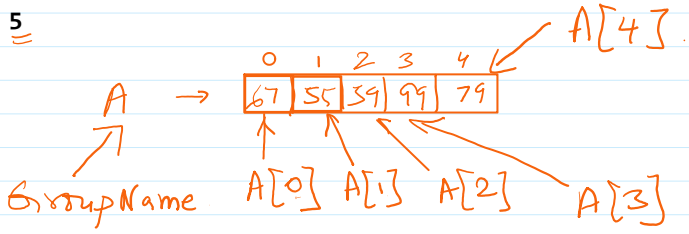
We can write very much flexible code using length property of Array:

```
import java.util.Scanner;
public class Two
{
    public static void input(Scanner sc, int A[],String msg){
        int i;
        System.out.print("\n"+msg+"\n");
```

```
            for(i = 0; i < A.length; i++){
                System.out.print("Enter value in ["+i+"]: ");
                A[i] = sc.nextInt();
            }
        }
        public static void printArray(int A[], String msg){
            int i;
            System.out.print("\n"+msg+"\n[");
            for(i = 0; i < A.length - 1; i++){
                System.out.print(A[i]+", ");
            }
            System.out.print(A[i]+"]\n");
        }
        public static void main(String args[]){
            Scanner sc = new Scanner(System.in);
            System.out.print("\fHow many elements you want: ");
            int N = sc.nextInt();
            int A[] = new int[N];
            input(sc, A,"\nEnter values in Array A[] \n");
            printArray(A,"\nArray A[]:");
        }
    }
```

WAP in Java to create an array of size N and print the maximum and minimum values from it. For example:

```
Enter number of elements: 5
[0]: 67
[1]: 55
[2]: 59
[3]: 99
[4]: 79
Output:
Maximum: 99
Minimum: 55
```




```
max = A[0];
min = A[0];
```

Keeps the count of elements present in the Array.

```
N =
int A[] = new int[N];
input(A, msg, sc);
print(A, msg);
printMaxAndMin(A);

for(i=1; i< A.length; i++){
    if (max < A[i]){
        max = A[i];
    }
    if (min > A[i]){
        min = A[i];
    }
}
```

```
package ArraysInJava;
import java.util.Scanner;

/**
 * Write a description of class Three here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Three
{
    public static void inputArray(int M[], String msg, Scanner sc){
        System.out.print("\n"+msg+"\n");
```

```
        int i;
        for(i = 0; i < M.length; i++){
            System.out.print("["+i+"]: ");
            M[i] = sc.nextInt();
        }
    }

    public static void printArray(int M[], String msg){
        System.out.print("\n"+msg+"\n[");
        int i;
        for(i = 0; i < M.length-1; i++){
            System.out.print(M[i]+", ");
        }
        System.out.print(M[i]+"]\n");
    }

    public static void findMaxAndMinInArray(int M[]){
        int i, max = M[0], min = M[0];
        for(i = 1; i < M.length; i++){
            if (max < M[i]){
                max = M[i];
            }

            if (min > M[i]){
                min = M[i];
            }
        }
        System.out.print("\nMaximum = " + max +
            "\nminimum = " + min);
    }

    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int N, A[];
        System.out.print("\fEnter number of elements: ");
        N = sc.nextInt();
        //Allocating memory for array
        A = new int[N]; //N- number of cells allocated in the heap memory.
        //call by reference
        inputArray(A, "Input values in A[]", sc); //delegating task to another function
        printArray(A,"Array A[]:");
        findMaxAndMinInArray(A);
    }
}
```

Output:

```
BlueJ: Terminal Window - AaditClass7
Options
Enter number of elements: 5

Input values in A[]
[0]: 23
[1]: 51
[2]: 6
[3]: 7
[4]: 19

Array A[]:
[23, 51, 6, 7, 19]

Maximum = 51
minimum = 6
```

WAP to create an array of size N, and print all the numbers which are either divisible by 2 or 3.
For example:
Enter number of elements: 5

```
Input values in A[]
[0]: 23
[1]: 51
[2]: 6
[3]: 7
[4]: 19

Array A[]:
[23, 51, 6, 7, 19]

Numbers divisible by 2 or 3
[1]: 51 divisible by 3
[2]: 6 divisible by 2
```

Home Work:

VII. Unsolved Programs on Single and Double Dimensional Arrays:

1. Write a program in Java to store 20 numbers (even and odd numbers) in a Single Dimensional Array (SDA). Calculate and display the sum of all even numbers and all odd numbers separately.
2. Write a program in Java to store 20 temperatures in °F in a Single Dimensional Array (SDA) and display all the temperatures after converting them into °C.

   Hint: $\frac{c}{5} = \frac{f - 32}{9}$
3. Write a program in Java to store 10 numbers (including positive and negative numbers) in a Single Dimensional Array (SDA). Display all the negative numbers followed by the positive numbers.

   Sample Input:

n[0]	n[1]	n[2]	n[3]	n[4]	n[5]	n[6]	n[7]	n[8]	n[9]
15	21	-32	-41	54	61	71	-19	-44	52

   Sample Output: –32, –41, –19, –44, 15, 21, 54, 61, 71, 52
4. Write a program in Java to store 20 numbers in a Single Dimensional Array (SDA). Display the numbers which are prime.

   Sample Input:

n[0]	n[1]	n[2]	n[3]	n[4]	n[5]	n[6]	n[7]	n[8]	n[9]
45	65	77	71	90	67	82	19	31	52

   Sample Output: 71, 67, 19, 31
5. Write a program to accept name and total marks of N number of students in two single subscript arrays name[ ] and totalmarks[ ].

   Calculate and print:

   (*a*) The average of the total marks obtained by N number of students.
   [average = (sum of total marks of all the students)/N]

   (*b*) Deviation of each student's total marks with the average.
   [deviation = total marks of a student – average] [ICSE 2018]